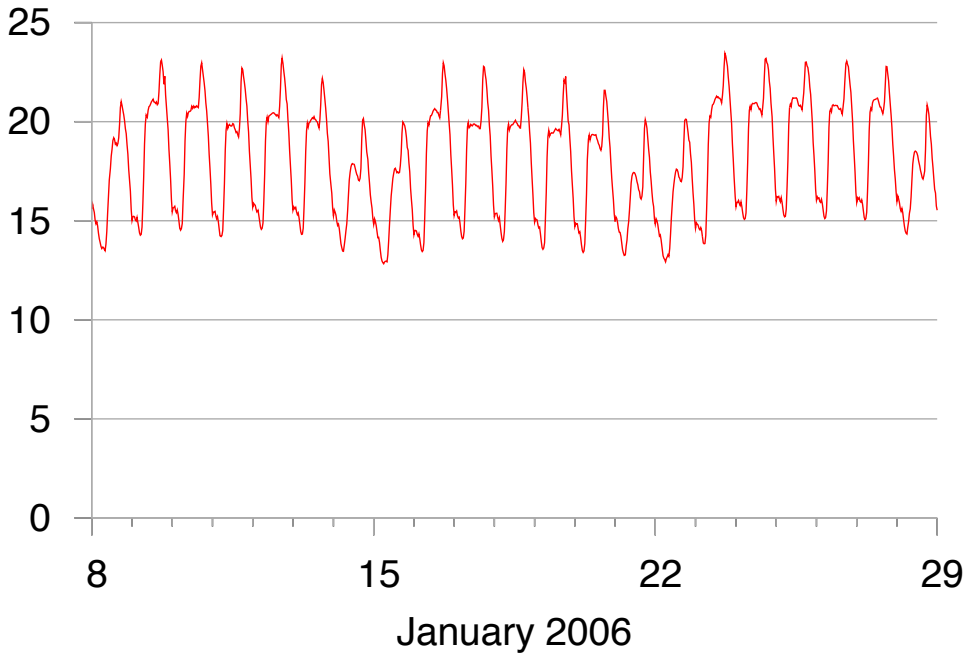
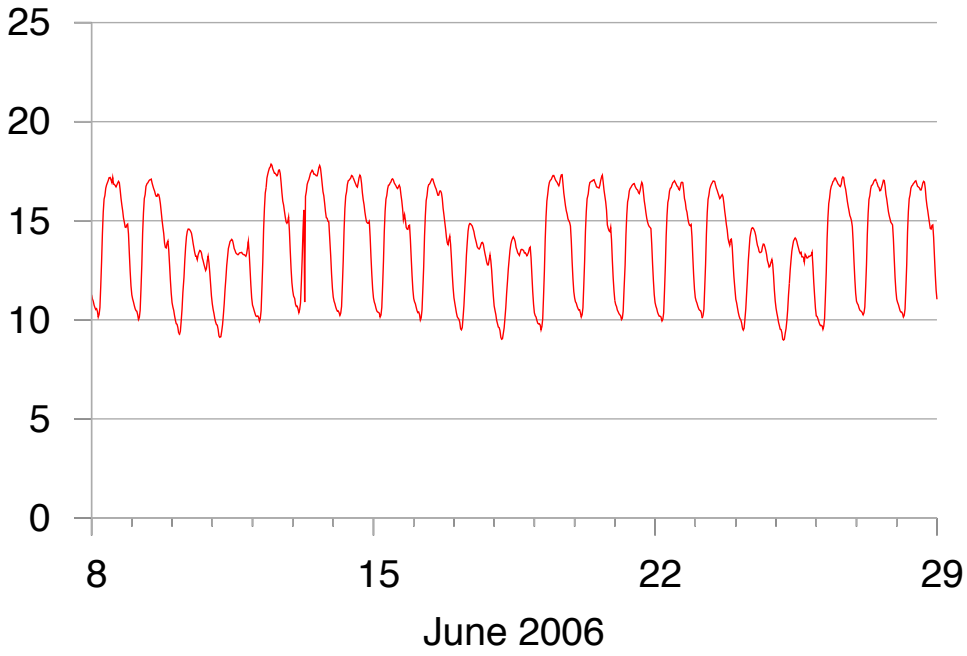
Problem Based Learning Session 2

- https://www.gridwatch.templar.co.uk
- The storage problem

Fluctuations in Electricity Demand

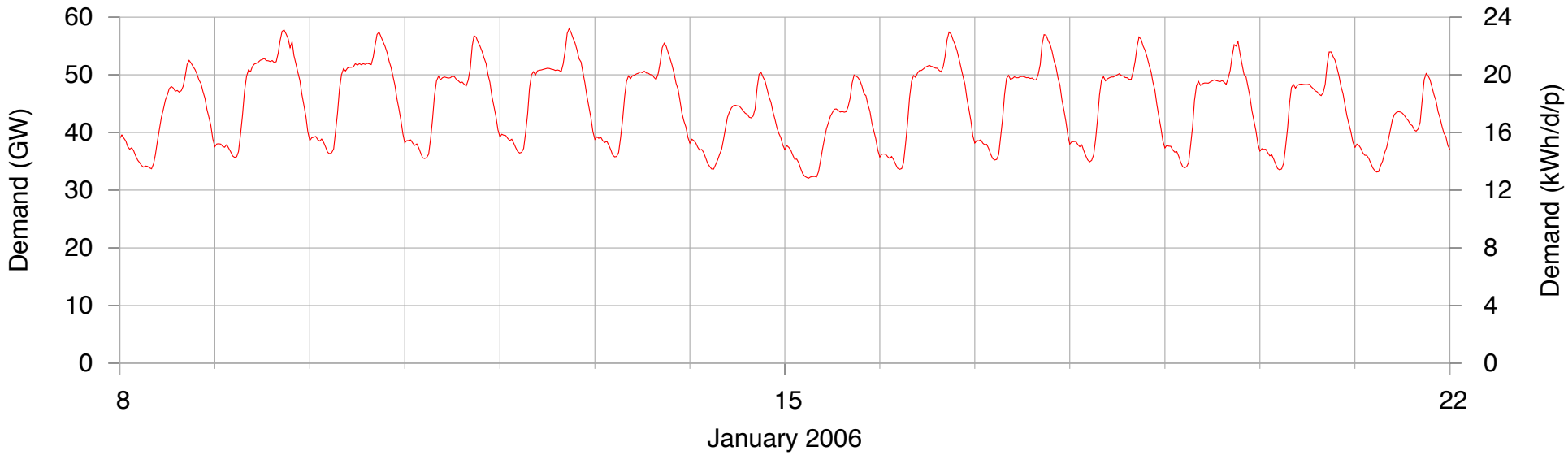
- UK electricity demand in kWh per day per person.
- Note daily, weekly and seasonal variation.





Fluctuations in Electricity Demand

- Zooming-in :

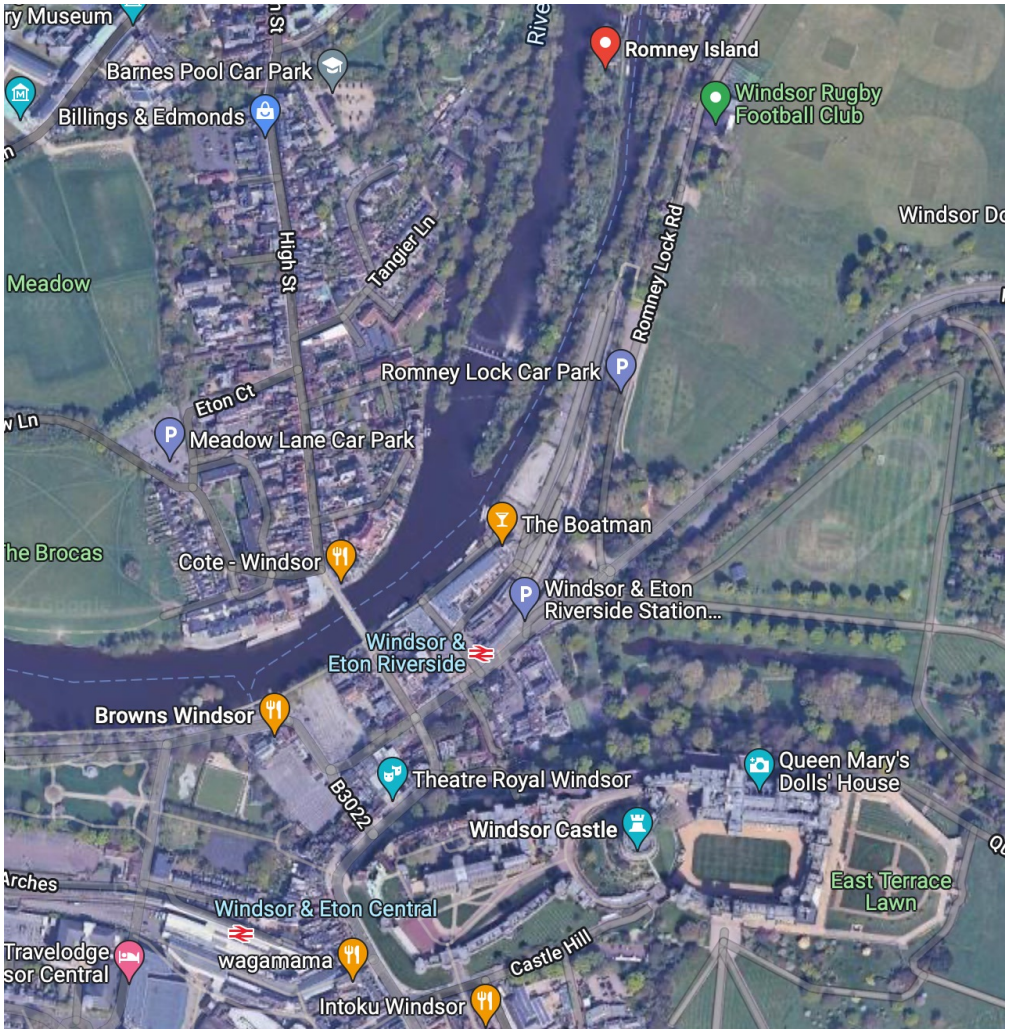


<u>Estimating the *slew-rate* :</u>

- 13 GW increase in demand between 6:30am and 8:30am.
- 6.5 GW/hour slew rate.

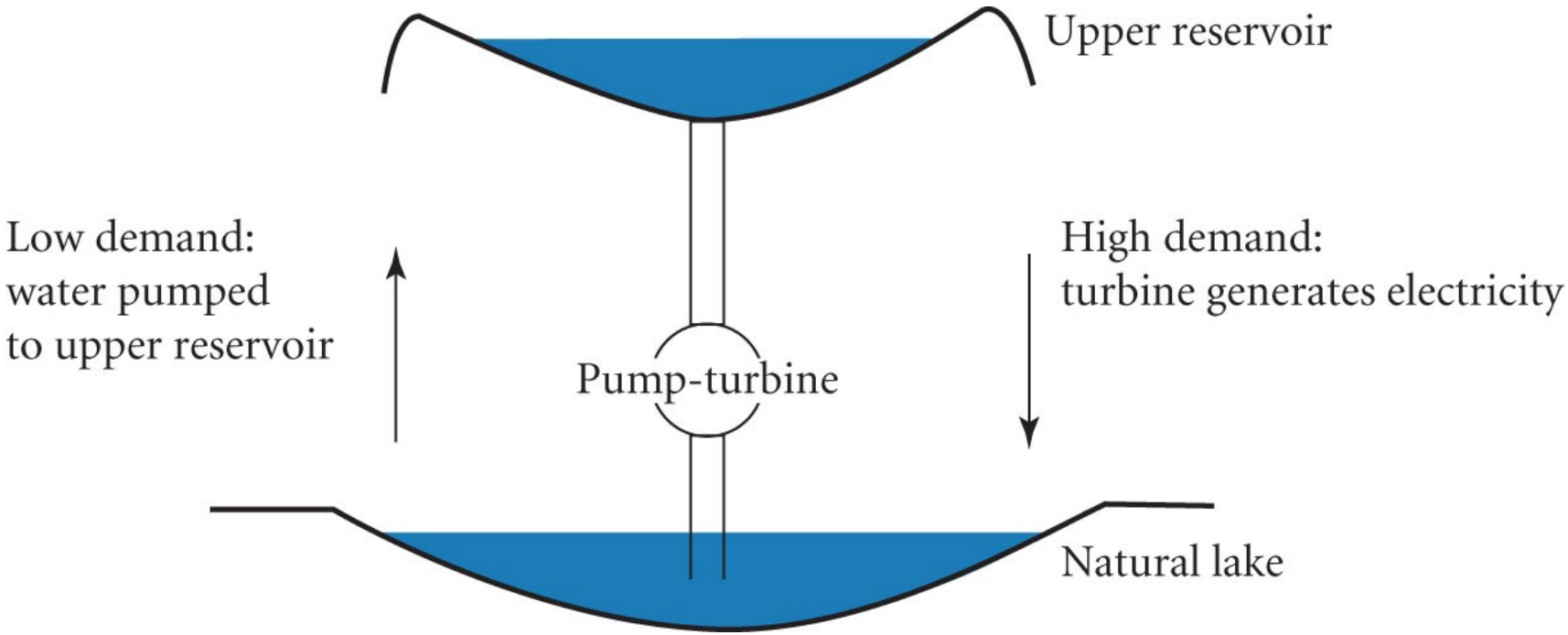
Problem Based Learning Session 3

PHAS0050 : Climate & Energy



Romney Weir hydro-electric scheme.

Pumped Storage



Will cover some of these aspects in more detail in hydro section:

- Potential energy of water stored in upper reservoir :

$$E = \rho g h V$$

- Power that can be extracted :

$$P = \eta \rho g h Q$$

V is the volume of reservoir

Q is the volume flow rate in $m^3\ s^{-1}$

h is the "head height"

Problem Based Learning Session 3

2016 Q9

(e) The Romney Weir hydroelectric project is located on a non-tidal stretch of the River Thames near Windsor Castle. 20% of the average river flow of 50 $m^3 s^{-1}$ is diverted through a generating station with a head of 2 metres. Neglecting any inefficiencies, make a simple estimate of the power generated. **[4]**

Given your result, comment on whether the capture of energy from non-tidal river flows can make a significant contribution to UK electricity generating requirements. **[2]**

Problem Based Learning Session 3

2017 Q9

(d) The UK plans to install a wind turbine generating capacity of 33 GW_e in order to supply an average of 10 GW_e. If a five day lull in the wind occurs how much energy in GWh needs to be stored in order to meet the shortfall? **[2]**

Compare this number to the typical capacity of a large pumped storage facility such as Dinorwig, capable of storing 6.7 million cubic metres of water at a head height of 500 m. Is pumped storage alone a feasible solution to managing wind power variability ? **[4]**

Problem Based Learning Session 3

2017 Q9

Consider a fleet of electric vehicles, each capable of holding batteries storing 40 kWh. How large would the fleet of vehicles need to be, in order to have a total electrical storage capacity equal to the wind shortfall calculated above ? Explain briefly how such an electric vehicle infrastructure might be used to help manage intermittent supply from renewables. [**4**]

Briefly explain the concept of a "smart grid" for managing fluctuations in electricity supply. [**2**]

Problem Based Learning Session 3

2017 Q9

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